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CERVICAL SPINE

Epidemiology of Cervical Injuries in NCAA Football Players

Andrew S. Chung, DO,* Justin L. Makovicka, MD,* Jeffrey D. Hassebrock, MD,* Karan A. Patel, MD,* Sailesh V. Tummala, BS,† David G. Deckey, BS,‡ Thomas C. Hydrick, BS,§ Nicolas C. Rubel, BS,§ and Anikar Chhabra, MD*

Study Design. Descriptive epidemiology study.

Objective. The purpose of this study was to describe the epidemiology of cervical spine injuries in collegiate football players.

Summary of Background Data. The incidence and etiology of cervical spine injuries in National Collegiate Athletic Association (NCAA) football players has not been well defined in recent years.

Methods. The incidence and characteristics of cervical spine injuries were identified utilizing the NCAA-ISP database. Rates of injury were calculated as the number of injuries divided by the total number of athlete-exposures (AEs). AEs were defined as any student participation in one NCAA-sanctioned practice or competition.

Results. An estimated 7496 cervical spine injuries were identified. Of these, 85.6% were categorized as new injuries. These occurred at a rate of 2.91 per 10000 AEs. Stingers were most common (1.87 per 10000 AEs) followed by cervical strains (0.80 per 10000 AEs). Injuries were nine times more likely to occur during competition when compared with practice settings. When compared with the regular season, the relative risks of sustaining a cervical spine injury during the preseason and postseason were 0.69 (95% CI 0.52–0.90) and 0.39 (95% CI 0.16–0.94), respectively. The rate of cervical spine injuries was highest in Division I athletes. Direct contact-related injuries were most common, representing 90.8% of all injuries sustained. Injuries were most common in linebackers (20.3%) followed secondarily by defensive linemen (18.2%). Most players returned to play within 24 hours of the initial injury (64.4%), while only 2.8% remained out of play for > 21 days.

Conclusion. Fortunately, the rate of significant and disabling cervical spine injuries appears to be low in the NCAA football athlete. The promotion of safer tackling techniques, appropriate modification of protective gear, and preventive rehabilitation in these aforementioned settings is of continued value.

Key words: athletes, cervical spine, cervical strain, collegiate football, contact sports, disc herniation, epidemiology, injury, NCAA, neck.

Level of Evidence: 4

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Approximately 70,000 student-athletes participate in National Collegiate Athletic Association (NCAA) football across three divisions.[1] While the overall risk of injury is arguably less in the collegiate football setting than at the professional level, the high-impact nature of the sport still predisposes athletes to injury.[2] In particular, young athletes who participate in contact sports are at highest risk for cervical spine injuries, as sports account for 15% of all cervical spine injuries.[3,4] The clinical severity of these injuries is highly variable, ranging from transient neuropraxias to life-threatening spinal cord injuries.

Unfortunately, while less common, severe cervical spine injuries have a higher propensity to be both clinically and personally devastating when compared with other sports-related injuries.[5,6] In a study evaluating severe injuries in all NCAA sports, head and neck injuries accounted for 11.2% of all injuries which resulted in extended time lost from play.[7] For NCAA athletes with aspirations of pursuing a professional career, a cervical spine injury can be career ending.[8]

Despite the substantial clinical and career-related implications of cervical spine injuries in the collegiate athlete, no recent characterization of these injuries exists. The purpose

From the *Department of Orthopedic Surgery, Mayo Clinic Arizona, Phoenix, AZ; †John A. Burns School of Medicine, University of Hawaii, Honolulu, HI; ‡Tufts University School of Medicine, Boston, MA; and §Mayo Clinic Arizona School of Medicine, Scottsdale, AZ.

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Address correspondence and reprint requests to Anikar Chhabra, MD, Department of Orthopaedic Surgery, Mayo Clinic Arizona, 5777 E. Mayo Blvd., Phoenix, AZ 85054; E-mail: Chhabra.anikar@mayo.edu

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of this study was to utilize a large NCAA database to update the current understanding regarding the epidemiology of cervical spine injuries in NCAA football athletes.

MATERIALS AND METHODS

Data Collection

The NCAA-Injury Surveillance Program (ISP) database was utilized to evaluate data from the 2009–2010 to 2013–2014 academic years. The NCAA-ISP is a prospectively gathered injury surveillance program managed by the Datalys Center for Sports Injury Research and Prevention, an independent nonprofit research organization. The study was found to be IRB exempt and approved by the research review board of the NCAA.

Use of the NCAA-ISP has been previously described in the literature[9,10] and consequently will only be briefly reviewed hereafter. The NCAA-ISP utilizes a voluntary convenience sample of NCAA football programs over a 5-year period. Consequently, there is variability in the number of programs participating in the data set each year.[9,11]As previously reported, this creates a deterministic sample of data, as opposed to a random sample, and has been used to monitor injury trends and patterns.[12]

Athletic trainers (ATs) at each participating program record injury and exposure data electronically through each institution's electronic health record. Data is collected during organized practices and competitions during the preseason, regular season, and postseason. For each injury, ATs and/or physicians complete a detailed report on the injury itself as well as the circumstances surrounding the injury. Injury data collected includes anatomic site of injury, diagnosis, circumstances of the injury, and event type. The date players return to participation is also recorded. Athletic trainers also record the number of student-athletes participating in each practice and competition to determine exposures.

The database was queried for football players in any division who sustained a "neck" or "cervical-spine" injury. This study relied upon the training and expertise of the ATs collecting data, as well as the other members of the medical staff assisting in documentation, to accurately diagnose and report all cervical spine injuries. The most recently updated diagnoses were used.

Computing National Estimates

To calculate national estimates of the number of cervical spine injuries, poststratification sample weights based on sport, division, and academic year were applied to each reported injury and athlete-exposures (AE). Poststratification sample weights were calculated with the following formula:

$$sample\ weight_{abc} = \left(\frac{number\ of\ teams\ participating\ in\ ISP_{abc}}{number\ of\ teams\ in\ NCAA_{abc}}\right)^{-1}$$

where $weight_{abc}$ is the weight for the ath sport of the bth division in the cth year. Weights for all data were further adjusted to correct for underreporting, accounting for the estimated 88.3% capture rate of all time-loss medical care injury events with the NCAA-ISP previously reported in the literature.[11,13] All numbers presented in this study are therefore estimates based off these calculations unless otherwise specified.

Data Analysis

The data was analyzed to assess rates and patterns of cervical-spine injuries sustained in collegiate athletes. All data analysis was performed utilizing weighted estimates as has previously been standardized in the literature.[12,14,15] Cervical-spine injuries were analyzed for injury type, time lost from participation in sports, time of season, event type, recurrence, injury mechanism, and participation restriction. The injury rate was defined as the number of injuries divided by the number of AEs. Athletic exposures were defined as any student-athlete participation in one NCAA-sanctioned practice or competition. The rates were reported as the ratio of injuries per 10000 AEs and calculated for an overall rate as well as individual rates for event type (practice *vs.* competition) and time of season (preseason, in-season, postseason).

Injury rate ratios were calculated to compare rates between event types and time of season, as they are useful for determining whether one participation type has an increased rate of injury compared with another. The following is an example of an injury rate ratio comparing injury rates between competition and practice:

$$IRR = \frac{\left(\frac{\Sigma\ Number\ of\ competition\ injuries}{\Sigma\ Competition\ AEs}\right)}{\left(\frac{\Sigma\ Number\ of\ practice\ injuries}{\Sigma\ Practice\ AEs}\right)}$$

All 95% confidence intervals (CI) not containing 1.0 were considered statistically significant. Participation restriction time was reported as intervals (<24 h, 1–6 d, 7–21 d, and >21 d) and descriptive data was presented as percentages of injuries. Data was analyzed using IBM SPSS software and Microsoft Excel.

RESULTS

A total of 300 injuries were identified in the database during the 2009–2010 to 2013–2014 academic years. This represented an estimated 7496 total cervical spine injuries over this time period. These occurred at a rate of 2.91 per 10000 AEs. Stingers were the most commonly identified injuries (1.87 per 10000 AEs) followed by cervical strains (0.80 per 10000 AEs) (Figure 1).

Injury by Event Type and Time of Season

Cervical spine injuries were approximately nine times more likely to occur during competition when compared with practice (Table 1). When compared with the regular season, the relative risks of sustaining a cervical spine injury

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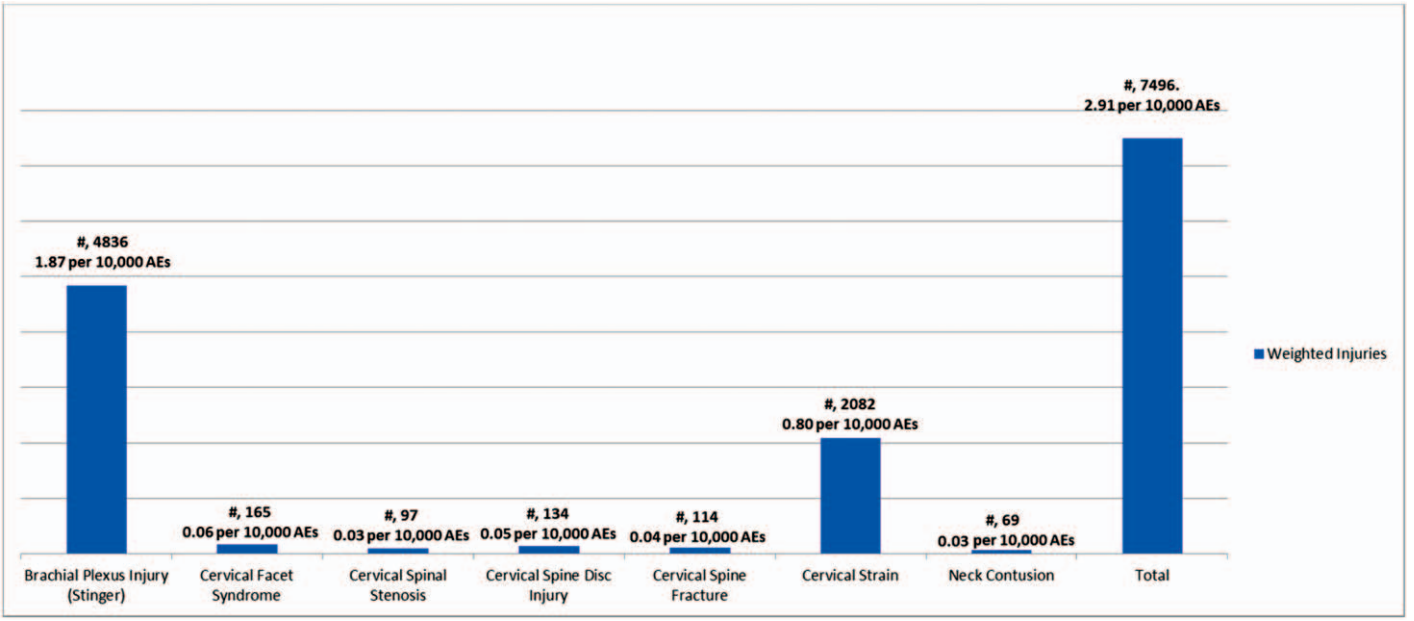

Figure 1. NCAAF National estimates for cervical spine injury types, totals, and rates: NCAA-ISP, 2009/2010-2013/2014 Academic Years.

during the preseason and postseason were 0.69 (95% CI 0.52–0.90) and 0.39 (95% CI 0.16–0.94), respectively (Table 2).

Injury by Collegiate Division and Athlete Position

The rate of cervical spine injuries was highest in Division I athletes (Figure 2). Injuries were most common in linebackers (20.3%) followed by defensive ends/tackles (18.2%). Notably, 100% of cervical spine fractures and 77.1% of cervical spinal stenosis occurred in linebackers. Quarterbacks were the least likely to sustain an injury of any type. Stingers were the most common injury sustained regardless of athlete position (Figure 3).

Mechanism of Injury, Recurrence Rates, Time Lost From Play, Need for Surgery

Contact-related injuries were most common representing 90.8% of all injuries sustained (Table 3). Of all injuries recorded, 85.6% were categorized as new injuries (Table 4). Most players returned to play within 24 hours of the initial

TABLE 1. Injury Occurrence in Relation to Practice and Competitions: NCAA-ISP, 2009/2010 to 2013/2014

Injury Type	Injury, n (%)*		Injury Rate per 10000 AEs		IRR† (95% CI)
	Practice	Competition	Practice	Competition	Competition/ Practice
Brachial plexus injury	2263 (59.8)	2573 (69.4)	0.97	10.28	10.57 (7.99–13.98)
Cervical facet syndrome	144 (3.8)	21 (0.6)	0.06	0.08	1.36 (0.16–11.60)
Cervical spinal stenosis	74 (2.0)	22 (0.6)	0.03	0.08	2.76 (0.25–30.48)
Cervical spine disc injury	134 (3.5)	0	0.05	—	—
Cervical spine fracture	0	114 (3.1)	—	0.45	—
Cervical strain	1123 (29.7)	959 (25.8)	0.48	3.83	7.94 (5.17–12.18)
Neck contusion	48 (1.3)	21 (0.6)	0.02	0.08	4.07 (0.37–44.85)
Total	3786	3710	1.62	14.82	9.11 (7.26–11.42)

*Percent of all injuries.
†IRR = injury risk ratio.
NCAA-ISP indicates National Collegiate Athletic Association-Injury Surveillance Program.

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TABLE 2. Distribution of Injuries in Relation to Season of Play: NCAA-ISP, 2009–2010 to 2013–2014

	Injury, n (%)*			Injury Rate per 10000 AEs			IRR† (95% CI)		
Injury Type	Preseason	Regular Season	Postseason	Preseason	Regular Season	Postseason	Post/ Preseason	Post/ Regular	Pre/ Regular
Brachial plexus injury	922 (50.9)	3872 (69.5)	43 (36.1)	1.13	2.29	0.46	0.40 (0.10–1.66)	0.20 (0.05–0.81)	0.50 (0.35–0.72)
Cervical facet syndrome	96 (5.3)	69 (1.2)	0	0.12	0.04	—	—	—	2.94 (0.59–14.58)
Cervical spinal stenosis	46 (2.5)	51 (0.9)	0	0.05	0.03	—	—	—	1.91 (0.17–21.05)
Cervical spine disc injury	48 (2.7)	87 (1.6)	0	0.06	0.05	—	—	—	1.17 (0.20–6.99)
Cervical spine fracture	0	114 (2.0)	0	—	0.06	—	—	—	—
Cervical strain	651 (35.9)	1354 (24.3)	76 (63.9)	0.81	0.80	0.81	1.00 (0.30–3.32)	1.02 (0.32–3.24)	1.02 (0.63–1.64)
Neck contusion	48 (2.7)	21 (0.4)	0	0.06	0.01	—	—	—	4.84 (0.44–53.35)
Total	1811	5568	119	2.27	3.30	1.27	0.56 (0.23–1.39)	0.39 (0.16–0.94)	0.69 (0.52–0.90)

*Percent of total injuries per time period.
†IRR = injury risk ratio.
NCAA-ISP indicates National Collegiate Athletic Association-Injury Surveillance Program.

injury (64.4%) while only 2.8% remained out of play for > 21 days (Table 5). Of all injuries reported, cervical disc issues were associated with the longest time lost from play (41.6%, 7–21 d; 29.2%, >21 d). None of the players in this study required surgery.

DISCUSSION

Cervical spine injuries are uncommon in the NCAA football athlete, occurring at a rate of 2.91 per 10000 AEs. Stingers were, by far, the most common injury sustained within the time period evaluated. Most injuries resulted in only transient

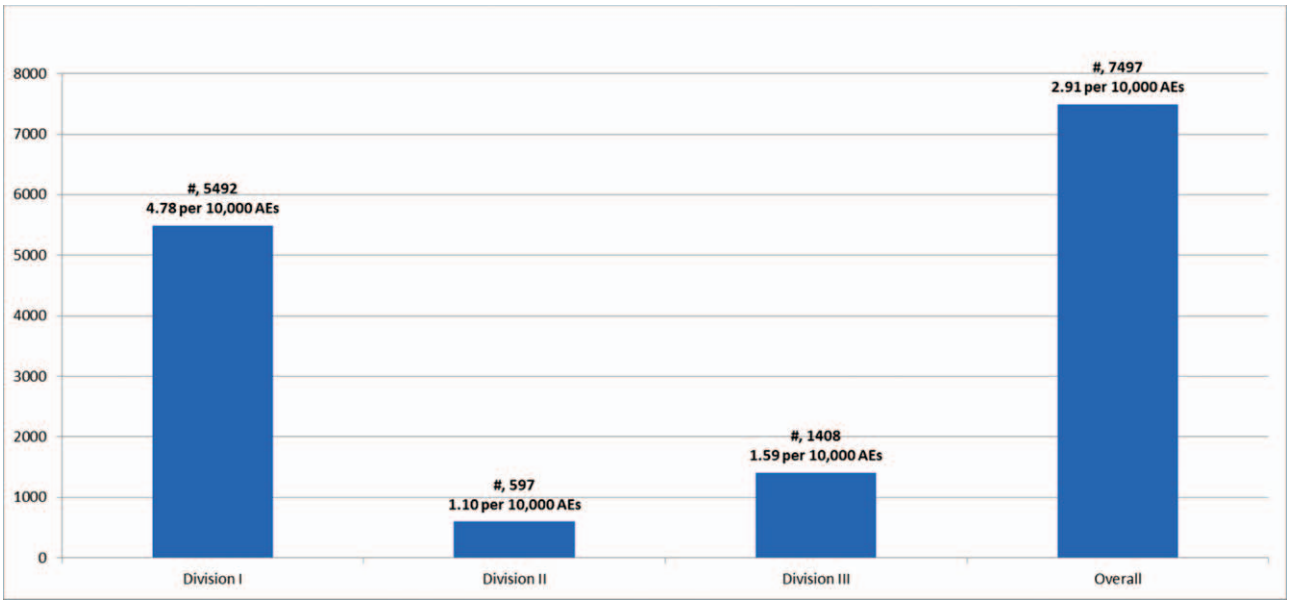

Figure 2. NCAAF injuries estimates by division: NCAA-ISP, 2009/2010-2013/2014.

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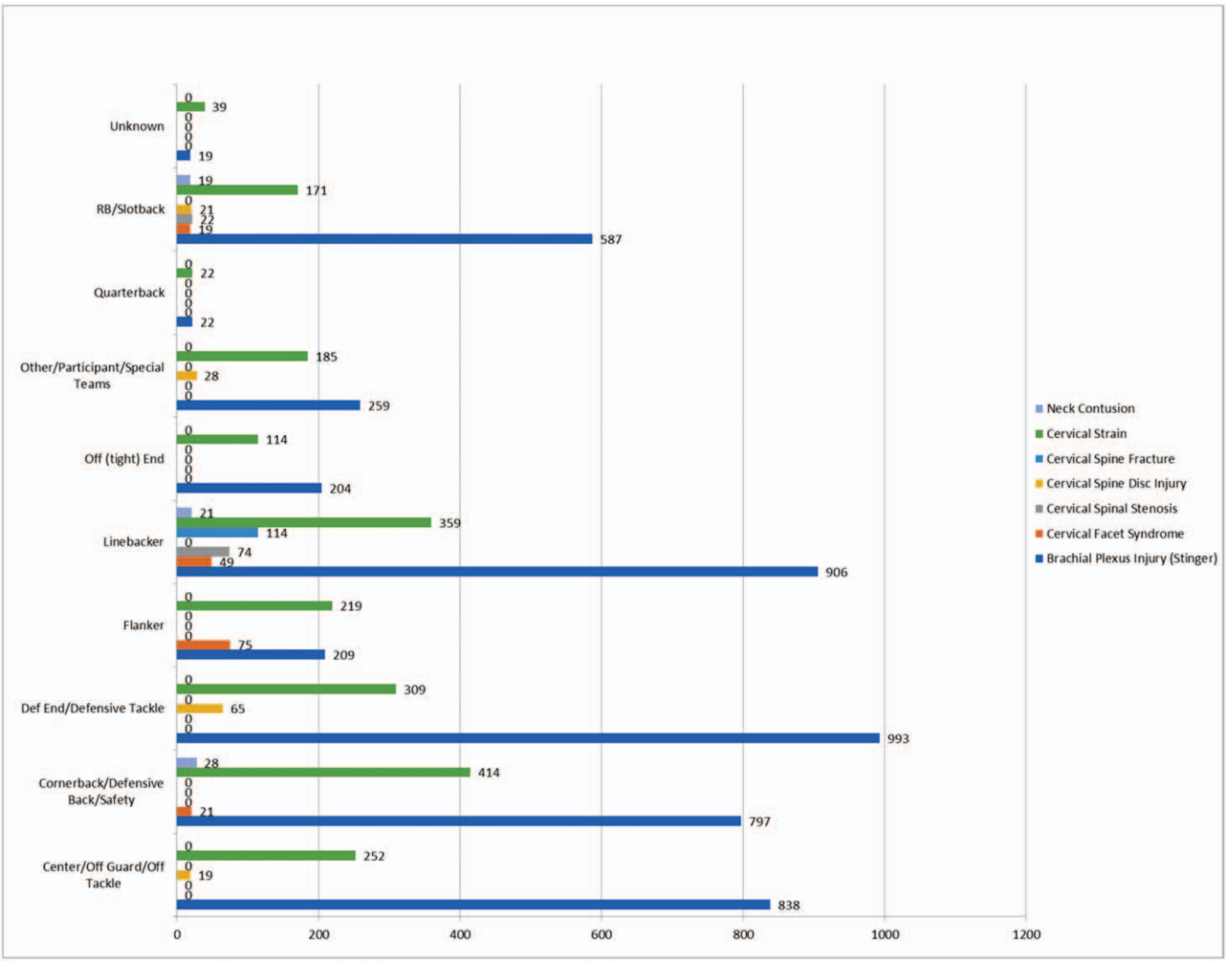

Figure 3. NCAAF Injuries by position: NCAA-ISP, 2009/2010-2013/2014.

time lost from play. Not surprisingly, cervical spine injuries most commonly occurred secondary to higher energy mechanisms of injury. Athletes were more likely to sustain injuries during the more rigorous regular season and almost entirely secondary to direct contact. Encouragingly, more severe injuries such as disc injury (1.8%) and fracture (1.5%) were rare and did not require surgical intervention, although they were much more likely to result in longer time lost from play.

Fortunately, more severe cervical spine injuries appear to be less common in the collegiate football athlete than the

TABLE 3. NCAAF Injury by Mechanism: NCAA-ISP, 2009–2010 to 2013–2014

Injury Type	Contact	Noncontact	Other	Overuse/ Gradual	Unknown	Total
Brachial plexus injury (stinger)	4595	66	0	39	137	4837
Cervical facet syndrome	110	0	0	0	55	165
Cervical spinal stenosis	51	0	0	0	46	97
Cervical spine disc injury	69	0	0	0	65	134
Cervical spine fracture	95	19	0	0	0	114
Cervical strain	1817	136	19	19	90	2081
Neck contusion	69	0	0	0	0	69
Total	6806	221	19	58	393	7497

NCAA-ISP indicates National Collegiate Athletic Association-Injury Surveillance Program.

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TABLE 4. NCAAF Recurrence of Injury by Injury Type: NCAA-ISP, 2009–2010 to 2013–2014

Injury Type	New Injury	Recurrent Injury Same Year	Recurrent Injury Previous Year	Total
Brachial plexus injury (stinger)	4002	616	218	4836
Cervical facet syndrome	165	0	0	165
Cervical spinal stenosis	97	0	0	97
Cervical spine disc injury	106	0	28	134
Cervical spine fracture	114	0	0	114
Cervical strain	1885	142	55	2082
Neck contusion	49	19	0	68
Total	6418	777	301	7496

NCAA-ISP indicates National Collegiate Athletic Association-Injury Surveillance Program.

professional football athlete.[16] Nonetheless, the potential implications of these cervical injuries on the athlete's career are immense. While most NCAA athletes do not progress to the professional level, football players with cervical spine pathology are less likely to be drafted to the National Football League than those without one. Furthermore, drafted players with cervical spine pathology have significantly shorter career lengths than players without.[17] Consequently, continued improvements in prevention strategies are warranted to help prevent these potentially career ending injuries.

Coaching staff and trainers should be aware that cervical spine injuries are most likely to occur during the regular season (74.3%). This is likely attributed to increased athlete effort during competitions, which translates into higher impact forces imparted by players in these situations. In this regard, it is not surprising that contact injuries, by far, are the most common etiology of cervical spine injury. This finding is consistent with existing literature in both the high school and collegiate setting.[18,19] While the NCAA-ISP data does not afford more specificity, axial loading, hyperextension, and multiple repetitive traumatic episodes have been reported as the most common mechanisms of contact injury.[20] These findings underscore the necessity for injury prevention programs that begin in the preseason, but perhaps most importantly, continue throughout the regular season.

The rate of cervical spine injuries was highest in Division I athletes. Fiercer competition and perhaps even stronger athletes may create a more injury-prone environment. Interestingly, quarterbacks were the least likely to sustain an injury of any type. This may be related to stricter targeting rules in NCAA football. On the contrary, and as has previously been reported in the literature, injuries were most common in linebackers (20.3%) followed secondarily by defensive linemen (18.2%); positions that require more repetitive tackling exposures.[6,7,16] Notably, 100% of cervical spine fractures and 77.1% of cervical spinal stenosis occurred in linebackers. As these injuries are likely the most consequential in terms of long-term disability, future efforts should focus on respective mitigation strategies.

This study is not without limitations. Data was reviewed in a retrospective manner which has its inherent limitations. Further, the NCAA-ISP, despite being a validated database, relies upon correct data-entry by the ATs, who are unfortunately not free from error. Additionally, there is potential for selection bias as participation in the database is entirely voluntary. Generalizability of this data to all NCAA football programs may consequently be limited. Finally, all analyses were done using weighted estimates which may have biased the statistical significance of the results. However, these limitations are accepted in exchange for the relatively large sample size available in the NCAA-ISP. As such, the authors believe that the data presented in this study offers the most current representation of the epidemiology of cervical spine injuries in the collegiate football athlete.

CONCLUSION

Stingers were the most common cervical spine injury sustained by NCAA football athletes. Fortunately, the rate of

TABLE 5. NCAAF Time Lost by Injury Type: NCAA-ISP, 2009–2010 to 2013–2014

	<24 h	1–6 d	7–21 d	>21 d	Total
Brachial plexus injury (stinger)	3115	724	344	111	4294
Cervical facet syndrome	124	40	0	0	164
Cervical spinal stenosis	0	46	22	0	68
Cervical spine disc injury	0	28	40	28	96
Cervical strain	900	761	107	40	1808
Neck contusion	49	19	0	0	68
Total	4188	1618	513	179	6498

NCAA-ISP indicates National Collegiate Athletic Association-Injury Surveillance Program.

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significant and disabling cervical spine injuries appears to be low. Injuries are most likely to occur during the regular season and secondary to direct contact. Furthermore, the linebacker and defensive linemen positions are at highest risk of injury. This information should be used to guide continued promotion of safer tackling techniques, appropriate modification of protective gear, and preventive rehabilitation in these aforementioned settings.

➢ Key Points

- ❑ The incidence and described mechanisms of cervical spine injury in the NCAA football athlete have not been described in recent years.
- ❑ The rate of significant and disabling cervical spine injuries appears to be low in the NCAA football athlete.
- ❑ The findings of this study can help to guide injury prevention strategies to target regular season practices for injury reduction.

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